

Understanding Boosting Through Margins

Exponential vs Logistic Loss

Decision Boundary Evolution

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Binary Classification Setup

Additive model:

$$F(x) = \sum_{m=1}^M \alpha_m h_m(x)$$

Prediction:

$$\hat{y} = \text{sign}(F(x))$$

Correct classification:

$$yF(x) > 0$$

Define margin:

$$\gamma(x) = yF(x)$$

From 0-1 Loss to Surrogate Loss

True error:

$$\mathbb{I}(yF < 0)$$

Upper bound:

$$\mathbb{I}(yF < 0) \leq e^{-yF}$$

Exponential loss:

$$L = e^{-yF}$$

Logistic loss:

$$L = \log(1 + e^{-yF})$$

Both depend only on margin.

Dataset

x	y
1	-1
2	-1
3	+1
4	-1
5	+1
6	+1

Best stump makes 2 errors.

How Splits Are Chosen

Candidate thresholds:

1.5, 2.5, 3.5, 4.5, 5.5

Exponential: Minimize weighted classification error.

Logistic: Minimize squared residual error:

$$\sum (r_i - \bar{r}_{region})^2$$

Iteration 1

Best split at 3.5

$$\epsilon_1 = \frac{2}{6} = 0.333$$

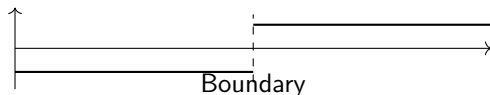
$$\alpha_1 = \frac{1}{2} \ln \frac{1 - \epsilon}{\epsilon} = 0.3466$$

$$F^{(1)} = \alpha_1 h_1$$

Decision Boundary After Iteration 1

Decision boundary defined by:

$$F^{(1)}(x) = 0$$



Single threshold at $x = 3.5$.

Margins After Iteration 1

x	$yF^{(1)}$
1	0.3466
2	0.3466
3	-0.3466
4	-0.3466
5	0.3466
6	0.3466

Two negative margins remain.

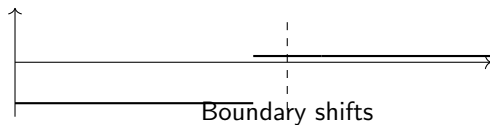
Iteration 2

Best split at 4.5

$$\alpha_2 = 0.255$$

$$F^{(2)} = \alpha_1 h_1 + \alpha_2 h_2$$

Decision Boundary After Iteration 2



Boundary moves right.

Margins After Iteration 2

x	$yF^{(2)}$
1	0.6016
2	0.6016
3	-0.0916
4	0.0916
5	0.0916
6	0.0916

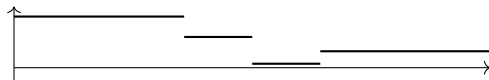
Only one negative margin.

Iteration 3

$$\alpha_3 = 0.15$$

All margins positive.

Decision Function After 3 Stumps



No zero crossing zero training error.

Loss:

$$L = \log(1 + e^{-yF})$$

Derivative:

$$\frac{\partial L}{\partial F} = -\frac{y}{1 + e^{yF}}$$

Pseudo-residual:

$$r_i = \frac{y_i}{1 + e^{y_i F}}$$

Iteration 1 — Logistic

Initialize:

$$F^{(0)} = 0$$

$$r_i^{(1)} = \frac{y_i}{2}$$

Split via SSE Minimization

At 3.5:

Left mean:

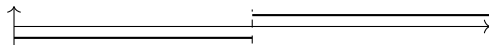
$$(-0.5 - 0.5 + 0.5)/3 = -0.167$$

Right mean:

$$(-0.5 + 0.5 + 0.5)/3 = 0.167$$

Chosen because it minimizes total SSE.

Decision Boundary (Logistic Iteration 1)



Smaller steps than exponential.

Boosting Advantage

Single stump: 66.7% accuracy

Boosted ensemble: 100% training accuracy

- Margins increase
- Loss decreases
- Decision boundary becomes flexible
- Weak learners combine to strong learner

Boosting performs:

Functional Gradient Descent

Sequentially correcting errors in function space.